

1 Sets

1.1 Introduction to Sets

A. Write each of the following sets by listing their elements between braces.

- 2. $\{3x + 2 : x \in \mathbb{Z}\} \rightarrow \{\dots, -7, -4, -1, 2, 5, 8, 11, \dots\}$
- 4. $\{x \in \mathbb{N} : -2 < x \leq 7\} \rightarrow \{1, 2, 3, 4, 5, 6, 7\}$
- 6. $\{x \in \mathbb{R} : x^2 = 9\} \rightarrow \{-3, 3\}$
- 8. $\{x \in \mathbb{R} : x^3 + 5x^2 = -6x\} \rightarrow \{-3, -2, 0\}$
- 10. $\{x \in \mathbb{R} : \cos x = 1\} \rightarrow \{2\pi x : x \in \mathbb{Z}\}$
- 12. $\{x \in \mathbb{Z} : |x| < 5\} \rightarrow \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$
- 14. $\{5x : x \in \mathbb{Z}, |2x| \leq 8\} \rightarrow \{-20, -15, -10, -5, 0, 5, 10, 15, 20\}$
- 16. $\{6a + 2b : a, b \in \mathbb{Z}\} \rightarrow \{2k : k \in \mathbb{Z}\} = \{\dots, -4, -2, 0, 2, 4, \dots\}$

B. Write each of the following sets in set-builder notation.

- 18. $\{0, 4, 16, 36, 64, 100, \dots\} \rightarrow \{4k^2 : k \in \mathbb{Z}\}$
- 20. $\{\dots, -8, -3, 2, 7, 12, 17, \dots\} \rightarrow \{7 - 5x : x \in \mathbb{Z}\}$
- 22. $\{3, 6, 11, 18, 27, 38, \dots\} \rightarrow \{n^2 + 2 : n \in \mathbb{N}\}$
- 24. $\{-4, -3, -2, -1, 0, 1, 2\} \rightarrow \{x \in \mathbb{Z} : -4 \leq x \leq 2\}$
- 26. $\{\dots, \frac{1}{27}, \frac{1}{9}, \frac{1}{3}, 1, 3, 9, 27, \dots\} \rightarrow \{3^x : x \in \mathbb{Z}\}$
- 28. $\{\dots, -\frac{3}{2}, -\frac{3}{4}, 0, \frac{3}{4}, \frac{3}{2}, \frac{9}{4}, 3, \frac{15}{4}, \frac{9}{2}, \dots\} \rightarrow \{\frac{3}{4}x - \frac{3}{4} : x \in \mathbb{Z}\}$

C. Find the following cardinalities.

30. $|\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\emptyset\}\}| = 5$
 32. $|\{\{\{1, 4\}, a, b, \{\{3, 4\}\}, \emptyset\}\}| = 1$
 34. $|\{x \in \mathbb{N} : |x| < 10\}| = 9$
 36. $|\{x \in \mathbb{N} : x^2 < 10\}| = 3$
 38. $|\{x \in \mathbb{N} : 5x \leq 20\}| = 4$

1.2 The Cartesian Product

A. Write out the indicated sets by listing their elements between braces.

2. Suppose $A = \{\pi, e, 0\}$ and $B = \{0, 1\}$.
- a) $A \times B = \{(\pi, 0), (\pi, 1), (e, 0), (e, 1), (0, 0), (0, 1)\}$
 - b) $B \times A = \{(0, \pi), (0, e), (0, 0), (1, \pi), (1, e), (1, 0)\}$
 - c) $A \times A = \{(\pi, \pi), (\pi, e), (\pi, 0), (e, \pi), (e, e), (e, 0), (0, \pi), (0, e), (0, 0)\}$
 - d) $B \times B = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$
 - e) $A \times \emptyset = \emptyset$
 - f) $(A \times B) \times B = \{$
 $((\pi, 0), 0), ((\pi, 1), 0), ((e, 0), 0), ((e, 1), 0),$
 $((0, 0), 0), ((0, 1), 0), ((\pi, 0), 1), ((\pi, 1), 1),$
 $((e, 0), 1), ((e, 1), 1), ((0, 0), 1), ((0, 1), 1)$
 $\}$
 - g) $A \times (B \times B) = \{$
 $(\pi, (0, 0)), (\pi, (0, 1)), (\pi, (1, 0)), (\pi, (1, 1)),$
 $(e, (0, 0)), (e, (0, 1)), (e, (1, 0)), (e, (1, 1)),$
 $(0, (0, 0)), (0, (0, 1)), (0, (1, 0)), (0, (1, 1))$
 $\}$
 - h) $A \times B \times B = \{$
 $(\pi, 0, 0), (\pi, 0, 1), (\pi, 1, 0), (\pi, 1, 1),$
 $(e, 0, 0), (e, 0, 1), (e, 1, 0), (e, 1, 1),$
 $(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1)$
 $\}$
4. $\{n \in \mathbb{Z} : 2 < n < 5\} \times \{n \in \mathbb{Z} : |n| = 5\} = \{(3, -5), (3, 5), (4, -5), (4, 5)\}$
6. $\{x \in \mathbb{R} : x^2 = x\} \times \{x \in \mathbb{N} : x^2 = x\} = \{(0, 1), (1, 1)\}$

$$8. \{0, 1\}^4 = \{ \\ (0, 0, 0, 0), (0, 0, 0, 1), (0, 0, 1, 0), (0, 0, 1, 1), \\ (0, 1, 0, 0), (0, 1, 0, 1), (0, 1, 1, 0), (0, 1, 1, 1), \\ (1, 0, 0, 0), (1, 0, 0, 1), (1, 0, 1, 0), (1, 0, 1, 1), \\ (1, 1, 0, 0), (1, 1, 0, 1), (1, 1, 1, 0), (1, 1, 1, 1) \\ \}$$

1.3 Subsets

A. List all the subsets of the following sets.

$$2. \{1, 2, \emptyset\} \rightarrow \{\}, \{1\}, \{2\}, \{\emptyset\}, \{1, 2\}, \{1, \emptyset\}, \{2, \emptyset\}, \{1, 2, \emptyset\}$$

$$4. \emptyset \rightarrow \{\}$$

$$6. \{\mathbb{R}, \mathbb{Q}, \mathbb{N}\} \rightarrow \{\}, \{\mathbb{R}\}, \{\mathbb{Q}\}, \{\mathbb{N}\}, \{\mathbb{R}, \mathbb{Q}\}, \{\mathbb{R}, \mathbb{N}\}, \{\mathbb{Q}, \mathbb{N}\}, \{\mathbb{R}, \mathbb{Q}, \mathbb{N}\}$$

$$8. \{\{0, 1\}, \{0, 1, \{2\}\}, \{0\}\} \rightarrow \{\}, \{\{0, 1\}\}, \{\{0, 1, \{2\}\}\}, \{\{0\}\}, \\ \{\{0, 1\}, \{0, 1, \{2\}\}\}, \{\{0, 1\}, \{0\}\}, \{\{0, 1, \{2\}\}, \{0\}\}, \{\{0, 1\}, \{0, 1, \{2\}\}, \{0\}\}$$

B. Write out the following sets by listing their elements between braces.

$$10. \{X \subseteq \mathbb{N} : |X| \leq 1\} \rightarrow \{\emptyset, \{1\}, \{2\}, \{3\}, \dots\}$$

$$12. \{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 1\} \rightarrow \{3\}, \{2\}, \{a\}$$

C. Decide if the following statements are true or false.

$$14. \mathbb{R}^2 \subseteq \mathbb{R}^3 \rightarrow \text{false}$$

$$16. \{(x, y) : x^2 - x = 0\} \subseteq \{(x, y) : x - 1 = 0\} \rightarrow \text{false}$$

1.4 Power Sets

A. Find the indicated sets.

$$2. \mathcal{P}(\{1, 2, 3, 4\}) = \{ \\ \emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \\ \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \\ \{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \\ \{1, 2, 3, 4\} \\ \}$$

$$4. \mathcal{P}(\{\mathbb{R}, \mathbb{Q}\}) = \{\emptyset, \{\mathbb{R}\}, \{\mathbb{Q}\}, \{\mathbb{R}, \mathbb{Q}\}\}$$

6. $\mathcal{P}(\{1, 2\}) \times \mathcal{P}(\{3\}) = \{$
 $(\emptyset, \emptyset), (\emptyset, \{3\}),$
 $(\{1\}, \emptyset), (\{1\}, \{3\}),$
 $(\{2\}, \emptyset), (\{2\}, \{3\}),$
 $(\{1, 2\}, \emptyset), (\{1, 2\}, \{3\})$
 $\}$
8. $\mathcal{P}(\{1, 2\} \times \{3\}) = \{\emptyset, \{(1, 3)\}, \{(2, 3)\}, \{(1, 3), (2, 3)\}\}$
10. $\{X \in \mathcal{P}(\{1, 2, 3\}) : |X| \leq 1\} = \{\emptyset, \{1\}, \{2\}, \{3\}\}$
12. $\{X \in \mathcal{P}(\{1, 2, 3\}) : 2 \in X\} = \{\{2\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}$

B. Suppose that $|A| = m$ and $|B| = n$. Find the following cardinalities.

14. $|\mathcal{P}(\mathcal{P}(A))| = 2^{2^m}$
16. $|\mathcal{P}(A) \times \mathcal{P}(B)| = 2^{m+n}$
18. $|\mathcal{P}(A \times \mathcal{P}(B))| = 2^{m \cdot 2^n}$
20. $|\{X \subseteq \mathcal{P}(A) : |X| \leq 1\}| = 2^m + 1$

1.5 Union, Intersection, Difference

2. Suppose $A = \{0, 2, 4, 6, 8\}$, $B = \{1, 3, 5, 7\}$ and $C = \{2, 8, 4\}$, Find:

- a) $A \cup B = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$
- b) $A \cap B = \emptyset$
- c) $A - B = A$
- d) $A - C = \{0, 6\}$
- e) $B - A = B$
- f) $A \cap C = C$
- g) $B \cap C = \emptyset$
- h) $B \cup C = \{1, 2, 3, 4, 5, 7, 8\}$
- i) $C - B = C$

4. Suppose $A = \{b, c, d\}$ and $B = \{a, b\}$. Find:

- a) $(A \times B) \cap (B \times B) = \{(b, a), (b, b)\}$
- b) $(A \times B) \cup (B \times B) = \{(b, a), (b, b), (c, a), (c, b), (d, a), (d, b), (a, a), (a, b)\}$

- c) $(A \times B) - (B \times B) = \{(c, a), (c, b), (d, a), (d, b)\}$
- d) $(A \cap B) \times A = \{(b, b), (b, c), (b, d)\}$
- e) $(A \times B) \cap B = \emptyset$
- f) $\mathcal{P}(A) \cap \mathcal{P}(B) = \{\emptyset, \{b\}\}$
- g) $\mathcal{P}(A) - \mathcal{P}(B) = \{\{c\}, \{d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{b, c, d\}\}$
- h) $\mathcal{P}(A \cap B) = \{\emptyset, \{b\}\}$
- i) $\mathcal{P}(A) \times \mathcal{P}(B) = \{$
 $(\emptyset, \emptyset), (\emptyset, \{a\}), (\emptyset, \{b\}), (\emptyset, \{a, b\}),$
 $(\{b\}, \emptyset), (\{b\}, \{a\}), (\{b\}, \{b\}), (\{b\}, \{a, b\}),$
 $(\{c\}, \emptyset), (\{c\}, \{a\}), (\{c\}, \{b\}), (\{c\}, \{a, b\}),$
 $(\{d\}, \emptyset), (\{d\}, \{a\}), (\{d\}, \{b\}), (\{d\}, \{a, b\}),$
 $(\{b, c\}, \emptyset), (\{b, c\}, \{a\}), (\{b, c\}, \{b\}), (\{b, c\}, \{a, b\}),$
 $(\{b, d\}, \emptyset), (\{b, d\}, \{a\}), (\{b, d\}, \{b\}), (\{b, d\}, \{a, b\}),$
 $(\{c, d\}, \emptyset), (\{c, d\}, \{a\}), (\{c, d\}, \{b\}), (\{c, d\}, \{a, b\}),$
 $(\{b, c, d\}, \emptyset), (\{b, c, d\}, \{a\}), (\{b, c, d\}, \{b\}), (\{b, c, d\}, \{a, b\})$
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10. Do you think the statement $(\mathbb{R} - \mathbb{Z}) \times \mathbb{N} = (\mathbb{R} \times \mathbb{N}) - (\mathbb{Z} \times \mathbb{N})$ is true, or false? Justify.

True. An ordered pair (x, n) lies in $(\mathbb{R} \times \mathbb{N}) - (\mathbb{Z} \times \mathbb{N})$ exactly when $(x, n) \in (\mathbb{R} \times \mathbb{N})$ but $(x, n) \notin (\mathbb{Z} \times \mathbb{N})$, that is, $x \in \mathbb{R}$ and $n \in \mathbb{N}$ but not both $x \in \mathbb{Z}$ and $n \in \mathbb{N}$. Since in both cases $n \in \mathbb{N}$, the second premise collapses to $x \notin \mathbb{Z}$, hence, the ordered pair satisfies $x \in (\mathbb{R} - \mathbb{Z})$ and $n \in \mathbb{N}$, that is, $(x, n) \in (\mathbb{R} - \mathbb{Z}) \times \mathbb{N}$, therefore, both sets have exactly the same elements.

1.6 Complement

2. Let $A = \{0, 2, 4, 6, 8\}$ and $B = \{1, 3, 5, 7\}$ have universal set $U = \{0, 1, 2, \dots, 8\}$. Find:

- a) $\bar{A} = B$
- b) $\bar{B} = A$
- c) $A \cap \bar{A} = \emptyset$
- d) $A \cup \bar{A} = U$
- e) $A - \bar{A} = A$
- f) $\overline{A \cup B} = \bar{U} = \emptyset$

g) $\bar{A} \cap \bar{B} = \emptyset$

h) $\overline{A \cap B} = U$

i) $\bar{A} \times B = B \times B = \{$
 $(1, 1), (1, 3), (1, 5), (1, 7),$
 $(3, 1), (3, 3), (3, 5), (3, 7),$
 $(5, 1), (5, 3), (5, 5), (5, 7),$
 $(7, 1), (7, 3), (7, 5), (7, 7)$
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1.7 Venn Diagrams

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1.8 Indexed Sets

2. Suppose $A_1 = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24\}$, $A_2 = \{0, 3, 6, 9, 12, 15, 18, 21, 24\}$, $A_3 = \{0, 4, 8, 12, 16, 20, 24\}$.

a) $\bigcup_{i=1}^3 A_i = \{0, 2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24\}$

b) $\bigcap_{i=1}^3 A_i = \{0, 12, 24\}$

4. For each $n \in \mathbb{N}$, **let** $A_n = \{-2n, 0, 2n\}$.

a) $\bigcup_{i \in \mathbb{N}} A_i = \{2n : n \in \mathbb{Z}\}$

b) $\bigcap_{i \in \mathbb{N}} A_i = \{0\}$

6.

a) $\bigcup_{i \in \mathbb{N}} [0, i + 1] = [0, \infty)$

b) $\bigcap_{i \in \mathbb{N}} [0, i + 1] = [0, 2]$

8.

a) $\bigcup_{\alpha \in \mathbb{R}} \{\alpha\} \times [0, 1] = \mathbb{R} \times [0, 1] = \{(x, y) : x \in \mathbb{R}, 0 \leq y \leq 1\}$

b) $\bigcap_{\alpha \in \mathbb{R}} \{\alpha\} \times [0, 1] = \emptyset$

10.

a) $\bigcup_{x \in [0,1]} [x, 1] \times [0, x^2] = \{(u, v) : 0 \leq u \leq 1, 0 \leq v \leq u^2\}$

b) $\bigcap_{x \in [0,1]} [x, 1] \times [0, x^2] = \{(1, 0)\}$

12. If $\bigcap_{\alpha \in I} A_\alpha = \bigcup_{\alpha \in I} A_\alpha$, what do you think can be said about the relationships between the sets A_α ?

They must all be equal.

14. If $J \neq \emptyset$ and $J \subseteq I$, does it follow that $\bigcap_{\alpha \in I} A_\alpha \subseteq \bigcap_{\alpha \in J} A_\alpha$? Explain.

Yes. If we let $x \in A_\alpha$ for every $\alpha \in I$, and since $J \subseteq I$, then, $x \in A_\alpha$ for every $\alpha \in J$, so it follows that $x \in \bigcap_{\alpha \in J} A_\alpha$. Therefore $\bigcap_{\alpha \in I} A_\alpha \subseteq \bigcap_{\alpha \in J} A_\alpha$.